Pre-filled: Using Laplace Transform to solve PDEs

Saturday, July 26, 2025 3:43 PM

10 - Use Laplace Transform to solve PDEs

10.1 Recall:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = F(s) \qquad \mathcal{L}^{-1}\{F(s)\} = f(t)$$

Table 1

$f(t)$	$F(s)$
1 or $u(t)$	$\frac{1}{s}$, $s > 0$
e^{at}	$\frac{1}{s-a}$, $s > a$
t^n, $n \epsilon \mathbb{N}$	$\frac{n!}{s^{n+1}}$, $s > 0$
$\sin at$	$\frac{a}{s^2+a^2}$, $s > 0$
$\cos at$	$\frac{s}{s^2+a^2}$, $s > 0$
$e^{at}\sin bt$	$\frac{b}{(s-a)^2+b^2}$, $s > a$
$e^{at}\cos bt$	$\frac{s-a}{(s-a)^2+b^2}$, $s > a$
$t^n e^{at}$, $n\epsilon\mathbb{N}$	$\frac{n!}{(s-a)^{n+1}}$, $s > a$
$u_c(t)$	$\frac{e^{-cs}}{s}$, $s > 0$
$u_c(t)f(t-c)$	$e^{-cs}F(s)$
$e^{ct}f(t)$	$F(s-c)$
$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$, $c > 0$
$\delta(t)$	1
$\delta(t-c)$	e^{-cs}
$f^{(n)}(t)$	$s^nF(s) - s^{n-1}f(0) - \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$

10.2 A physical Example

Semi-Infinite String

Find the displacement $w(x, t)$ of an elastic string subject to the following conditions. (We write w since we need u to denote the unit step function.)

(i) The string is initially at rest on the x-axis from $x = 0$ to ∞ (*"semi-infinite string"*).

(ii) For $t > 0$ the left end of the string ($x = 0$) is moved in a given fashion, namely, according to a single sine wave

$$w(0,t) = f(t) = \begin{cases} \sin t & \text{if } 0 \leqq t \leqq 2\pi \\ 0 & \text{otherwise} \end{cases} \qquad \text{(Fig. 316).}$$

(iii) Furthermore, $\lim_{x\to\infty} w(x,t) = 0$ for $t \geqq 0$.

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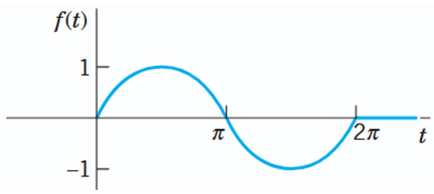

Fig. 316. Motion of the left end of the string in Example 1 as a function of time t

See: battle rope exercise: [How to perform BATTLE ROPES - HOIST Fitness MotionCage Exercise]

10.3 The <u>Problem</u>:

Wave equation: $\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2}$ for $x > 0$, $t > 0$

BC: $w(0,t) = f(t) = \begin{cases} \sin(t), & 0 \le t \le 2\pi \\ 0 & \text{otherwise} \end{cases}$; $\lim_{x \to \infty} w(x,t) = 0$

IC: $w(x,0) = 0$, $w_t(x,0) = 0$

10.4 Use <u>Laplace Transform</u> on both sides of the PDE, given initial conditions.

10.5 Solving the DE: Similar to $W'' - \frac{s^2}{c^2}W = 0 \Rightarrow W = Ae^{sx/c} + Be^{-sx/c}$

(use the boundary conditions

10.6 Apply inverse Laplace:

Since $\mathcal{L}^{-1}\{F(s)\,e^{-as}\} = f(t-a)\,u(t-a)$

in this case:

This is a single sine wave traveling to the right with speed c.

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